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QUESTION BANK

Semester – III Examinatation – 2026

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MAJ-4

ITES & IOT

PYTHON IN ITES & IOT

UNIT I – List, Ranges & Tuples in Python

2 Marks

1. What is a List in Python?
2. What is a Tuple in Python?
3. What is List Comprehension?
4. What is Nested List Comprehension?
5. What is an Iterator in Python?
6. What is a Generator in Python?
7. What is the use of yield in Python?
8. What is a Lambda Function?
9. What is the use of range() function?
10. What is the difference between range() and next()?
11. What are Ordered Sets?
12. What is the difference between List and Tuple?
13. What is the use of next() function?
14. What are Comprehensions in Python?

5 Marks

1. Explain List Comprehension in Python. Describe how it improves code readability and performance, with examples of filtering and transforming data.
2. Discuss various built-in methods available for Lists and Tuples in Python with examples.
3. What is a Lambda Function in Python? Explain its syntax, use cases and limitations with examples involving sorting and filtering.
4. Explain Lists, Ranges and Tuples in Python with suitable examples.
5. Explain Nested List Comprehensions with examples.
6. Explain Iterators and Generators in Python.
7. Explain the yield and next() functions with examples.
8. Explain Ordered Sets with suitable examples.
9. Explain the use of Lambda Expressions in Python.
10. Differentiate between List, Tuple and Set.

UNIT II – Usage in IoT and ITES

2 Marks

1. What are Desktop Applications?
2. What is Tkinter?
3. What is PySide?
4. Mention any two uses of PySide.
5. State one advantage of using PySide over other GUI frameworks.
6. What are Web Applications?
7. What is Scientific Computing?
8. What are IoT Applications?
9. What are Mobile Applications?
10. What is Data Analytics?
11. What is Data Processing?
12. Mention two applications of Python in ITES.

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5 Marks

1. Write a short note on Desktop Applications of IoT.
2. What is your understanding about ITES?
3. Explain the role of Python in Desktop Applications and Web Applications.
4. Discuss Event-Driven Programming in GUI Frameworks. Explain how Signals and Slots work in PySide with examples.
5. Explain the uses of Tkinter and PySide in Python.
6. Explain Scientific Computing with examples.
7. Discuss the applications of Python in IoT and Mobile Applications.
8. Explain the role of Data Analytics and Processing in ITES.
9. Differentiate between Desktop Applications and Web Applications.

UNIT III – Introduction to IoT

2 Marks

1. What is Internet of Things (IoT)?
2. What is the history of IoT?
3. What are the basic requirements of IoT?
4. Mention any two functions of IoT.
5. What is IoT Architecture?
6. What are IoT Enabling Technologies?
7. What is the Network Layer in IoT?
8. What is the Application Layer in IoT?
9. What is Middleware Layer in IoT?
10. What is End-to-End Communication in IoT?
11. What is Wired Communication?
12. What is Wireless Communication?
13. Mention two major Hardware Components of IoT.
14. Mention two Enabling Technologies of IoT.
15. What is the role of Sensors in IoT?
16. What is the role of Actuators in IoT?

5 Marks

1. What are the major applications of IoT? Discuss in detail.
2. Discuss the birth of Internet. Define Internet of Things (IoT) in your own words.
3. Critically describe the major components of IoT.
4. Discuss the important Hardware Components used in IoT systems such as Sensors, Actuators, Microcontrollers, Gateways and Communication Modules, with examples.
5. Describe IoT Enabling Technologies with examples.
6. Differentiate between Network Layer and Application Layer in IoT.
7. Explain the functions and structure of IoT.
8. Explain IoT Architecture with a suitable diagram.
9. Explain the role of Wired and Wireless Communication in IoT.
10. Explain the different Communication and Networking Protocols used in IoT.
11. Explain the requirements and characteristics of IoT.
12. Discuss the role of Middleware in IoT Architecture.

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UNIT IV – IoT Case Study

2 Marks

1. What is an IoT Case Study?
2. What are Project Requirements?
3. What is Hardware Requirement?
4. What is Software Requirement?
5. What is IoT Application Design?
6. What is a Web-based IoT Application?
7. What is a Mobile-based IoT Application?
8. What is a Device-based IoT Application?
9. What is a Data Generator?
10. What is a Minimum Project?
11. What is meant by Projects on Every Technology?

5 Marks

1. How to design an IoT Application? Explain briefly.
2. Explain the steps involved in an IoT Project Case Study.
3. Discuss Hardware and Software Requirements for an IoT Project.
4. Explain the design of Web, Mobile and Device-based IoT Applications.
5. Explain the importance of Project Requirements in IoT.
6. Discuss the role of Data Generators in IoT Projects.
7. Explain the steps involved in developing a Minimum IoT Project.
8. Describe the implementation process of an IoT Project with an example.

UNIT V – Combining IoT and ITES (Enabling IoT Devices)

2 Marks

1. What is Cloud Storage?
2. What is Server Storage?
3. What is In-house Storage?
4. What is Database Connectivity with IoT?
5. What is MySQL?
6. What is Data Analysis?
7. What is Data Visualization?
8. What is Data Interpretation?
9. What is Data Cleaning?
10. What is the role of Python Modules in Data Analysis?
11. What is Final Project Implementation?
12. Mention two advantages of Cloud Storage.

5 Marks

1. Discuss the Cloud Computing Industry and its growth in India.
2. Write a short note on Interpretation of Data in IoT.
3. Explain the role of Cloud Computing in IoT Architecture. Discuss data storage, analytics, dashboards and remote access.
4. Describe Edge Computing and its relevance in IoT. Highlight the differences between Edge, Fog and Cloud Computing with examples.
5. Explain the different types of Storage used in IoT.
6. Explain Database Connectivity with IoT using MySQL.
7. Explain the uses and importance of Database Connectivity in IoT.
8. Explain Data Analysis using Python Modules.
9. Discuss Data Visualization and Data Interpretation in IoT.
10. Explain Data Cleaning in IoT.
11. Describe the steps involved in Final Project Implementation.

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